

Vikramaditya Singh

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EDUCATION

IIIT Guwahati

B. Tech 2017-2021

Computer Science Engineering
Cum. GPA: 8.48

Modern Delhi Public School

12th Science from CBSE , May 2016

Physics(A1), Maths (A1),

Chemistry(A2) and 87% (Overall)

Dynasty International School

10th from CBSE , May 2014

9.8 CGPA

SKILLS

Languages

Java, C, Python, Go, Perl,
Javascript

Frameworks

ReactJs, NextJs, Spring, Spring
Boot, Hibernate

Databases

MySQL, MariaDB, MongoDB, Redis
Cloud

AWS

Tools

Swagger, Postman, Nginx

Others

Docker, Jenkins, Git, Terraform,
Lucene

Methodologies

TDD, Agile

AI

Deep Learning, Machine Learning

PROFILE

Computer science engineering graduate with expertise in Algorithms, Data Structures, Application Development, Linux, Networking, Database and have good knowledge of Cyber-Security. I am a good listener, imaginative, eager to learn, improve, and highly adaptable. Reliability, ownership, and excellent interpersonal skills are my core qualities.

EXPERIENCE

CISCO

Software Engineer 2

August 2021 - present

Bangalore, India

- Improved the NAP & IPS policy snapshot performance reducing the operation time by 70%, which in turn reduced the time to deploy the configuration to the devices from the FMC.
- Designed & implemented the Zero trust policy on FMC which is a key feature of the product. Implemented the complete backend which included changes in the API, global search, database, service & model layer, various validations, reporting, audit & telemetry.
- Implemented the changes needed for deploying the Zero Trust policy to the device. Added the Snort3 configuration creation & handled the Lina CLI generation by enhancing the parser with multiple new commands.
- Added ability to auto-enroll the certificate selected within the Zero trust policy onto the device which greatly enhanced the user experience.
- Extracted the Rule Conflict Detection & Object overlap feature available for the Access Control Policy as a rule analyzer library and enhanced its logic. This library is used on both Firepower Management Console (FMC) & Cloud Defense Orchestrator (CDO)
- Added the expired rule detection & mergeable rule detection to the rule analyzer library.
- Created the complete CI/CD pipeline for the library on Jenkins & maintained its various release on the Jfrog artifactory.
- Created micro-service using the Rule Analyser library to provide analysis for the cdFMC enrolled in Cisco Defense Orchestrator. Worked with AWS services like S3, SQS, SNS & Elastic Container. This enabled the customer to optimise there policy configuration

COURSEWORK

Data Structure & Algorithms
Computer Network & Security
Operating System
RDBMS
OOP & Design Patterns
Software Engineering
Deep Learning

AWARDS

Dean's List

ACTIVITIES

Programming Club Coordinator

HACKATHON

Smart India Hackathon 2019
Runner up Chandigarh, India

Software Engineer 1

May 2020 – July 2020

- Improved the Firepower management console's device listing & management page performance, reduced the page load time by 96%.
- Added Elephant Flow Detection feature to the Access Policy by implementing the complete backend which included changes in the database schema, validations, models & service layer. Also did changes for deploying the configuration to the Lina & Snort3 on the device.
- Added pdf reporting, audit log, delta preview, telemetry functionality for the Elephant Flow Detection & Threat Detection Setting feature.
- Implemented dynamic warning framework for the device upgrade flow on the FMC using React.

SECURONIX

SDE Intern

April 2021 - August 2020

Bangalore, India

- Worked with spring framework on email microservice and automatic incident report creation using jasper report.
- Wrote various scripts for automating querying and updating tasks.
- Automated report creation for analysis from the data for CTA team.

CISCO

SDE Intern

May 2020 – July 2020

Bangalore, India

- Worked on Snort 2.9.16 & developed a gRPC detector and integrated it with the FMC console to enable generation of alerts.
- Worked on the preprocessor framework for scanning the message content transferred via. gRPC for malicious payloads.

PROJECTS

Domain Generating Algorithm Detection

A Domain Generating Algorithm (DGA) is a program or subroutine that provides malware with new domains on-demand or on the fly. Detected the algorithmically generated domain with an accuracy of 96% and classified the detected domains to its respective DGA variant by accuracy of 87% and was able to detect unknown DGA variants by using clustering.

Leave Management System

Built a complete leave management system using Flask and VueJs with features like email notification, visualising the leaves records etc. Created the SRS, ER, Data flow, Class, Sequence and Use case diagram. Followed the TDD approach and released it as a docker image.